pfi

# Python for Fixed Income

COURSE 7 · NLP, RAG, AND LLMS

## Text in Python: FOMC Statements and an Indenture

WEEK 1 · DAY 1

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# Text as data, in three weeks

### Week 1: Text as data

Strings, TF-IDF, a classifier, word vectors

### Week 2: Transformers on the CPU

Embeddings, search by meaning, EDGAR

### Week 3: LLMs, with checks

Covenant extraction checked against the text

### Today's two documents

**226 FOMC statements**

2000-02-02 to 2026-09-16, the Board's own text

**One indenture, from EDGAR**

Boyd Gaming Corporation, 4.750% Senior Notes due 2031: the teaching indenture for all three weeks

**Today: text as Python sees it. No model, no key.**

# Real documents, described

## An FOMC statement's output

| Allowed names | Never called |
| --- | --- |
| count of words from the course's list | hawkish or dovish |
| dictionary score | a signal or a policy view |
| the action the statement states | a forecast of rates |

## An indenture's terms

| Allowed names | Never called |
| --- | --- |
| the indenture states | loose, tight, weak, strong |
| extracted value, with its quote | covenant-lite or protective |
| passed the source check | a credit view or recommendation |

# Text is a string

**1,254**

characters in the first statement

**6**

text.count("rate")

**5**

the whole word "rate"

```
text[:300]                  # a slice: the first 300 characters
"Inflation" in text         # False: in is exact about capitals
words = text.split()        # pieces, with punctuation glued on
text.count("rate")          # 6: also inside 'generate'
```

**A substring count is not a word count, in Python or in Excel.**

226 statements, 213 after scheduled meetings and 13 unscheduled. Source: the Board of Governors, federalreserve.gov, one page per statement.

# Why the file starts in 2000

| | |
|---|---|
| **February 1994** | The FOMC first announced the outcome of a meeting. |
| **February 1995** | All changes in the stance of policy communicated immediately. |
| **January 2000** | A statement after each regularly scheduled meeting. |

**The Board's FAQ dates statements after every meeting from May 1999. The course quotes both.**

Sources, read 8 October 2026: federalreserve.gov/monetarypolicy/fomc_historical.htm (Transcripts and other historical materials); federalreserve.gov/faqs/federal-open-market-committee-fomc-not-public.htm

# Characters that look alike

| Code | Name | Times | Statements |
|---|---|---|---|
| U+2011 | NON-BREAKING HYPHEN | 50 | 34 |
| U+2013 | EN DASH | 3 | 3 |
| U+2014 | EM DASH | 2 | 1 |
| U+2019 | RIGHT SINGLE QUOTATION MARK | 25 | 19 |

### Searching "longer-term"

Plain hyphen: 102 statements. With U+2011: 7 more, missed. After normalize_text: 109.

```
textkit.normalize_text(t)    # NFC, the map, white space tidied: 0 characters outside ASCII left
```

**The file keeps the Board's characters. Normalizing happens in code, on both sides.**

# Regular expressions: the target range

```
NUMBER = r"(?:\d+[- ]\d+/\d+|\d+/\d+|\d+)"        # 5-3/4, 1 3/4, 1/4, 6
RANGE  = ...federal funds rate ... (?P<low>NUMBER) to (?P<high>NUMBER) percent
TARGET = ...target for the federal funds rate ... (?P<target>NUMBER) percent
```

**144**

state a range, from 2008-12-16

**76**

a single target, to 2008-10-29

**6**

neither: all unscheduled

**On the raw text: 132 ranges, 12 missed, every one written with U+2011.**

*A pattern is a claim about how a document is written. The counts test it.*

# Tokens

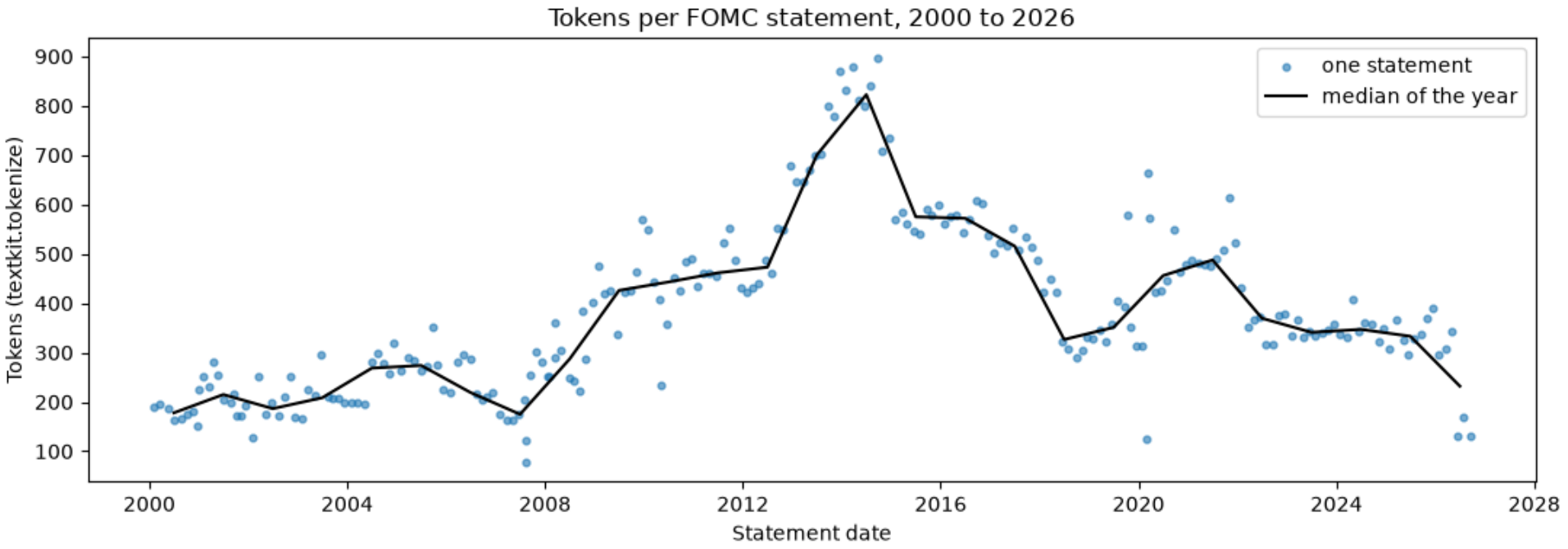
```
TOKEN_PATTERN = r"[a-z0-9]+(?:[-'.][a-z0-9]+)*"
tokenize("agency mortgage-backed securities")
  ['agency', 'mortgage-backed', 'securities']
tokenize("0 to 1/4 percent")
  ['0', 'to', '1', '4', 'percent']
```

First statement: 190 tokens against 188 split pieces. All statements: 86,576 tokens, 1,910 distinct; stop words are a share of 0.3776.

| Token | Count |
|---|---|
| committee | 1,601 |
| inflation | 1,415 |
| economic | 952 |
| federal | 869 |
| rate | 829 |
| policy | 755 |

**Tokens are for counting words. Patterns are for reading numbers.**

# The statements got longer




**A count describes each document by date. It does not say why the length changed.**

# A first look at an indenture

**63,595**

words in the indenture

**214**

heading lines for 107 sections

**198,454**

where the body's 4.12 starts

| Section | Title | Words |
|---|---|---|
| 4.10 | Change of Control | 1204 |
| 4.11 | Asset Sales | 1138 |
| 4.12 | Limitation on Indebtedness | 1819 |
| 4.14 | Limitation on Liens | 511 |
| 4.15 | Limitation on Restricted Payments | 1653 |
| 4.20 | Certain Suspended Covenants | 339 |

### Why the contents go

Every number matches twice. find_sections keeps the last heading: the body's.

Source: Boyd Gaming Corporation, indenture for the 4.750% Senior Notes due 2031, Exhibit 4.1 to the 8-K filed 2021-06-08, accession 0001193125-21-185495.

# Defined terms, and the ones the pattern misses

**136**

"Term" means, or has the meaning

**148**

lines that start with a quoted term

**12**

written another way: missed

### How the missed definitions are written

"... shall mean ...", "... shall be deemed to occur if ...", and "Consolidated Total Assets" of any Person as of any date means ...

**Test every pattern on the document it runs on. Day 12 widens this one.**

Source: Boyd Gaming Corporation, indenture for the 4.750% Senior Notes due 2031, Exhibit 4.1 to the 8-K filed 2021-06-08, accession 0001193125-21-185495, Section 1.01.

# Stemming, named and not used

| Token | Count |
|---|---|
| Restricted | 412 |
| restrictions | 43 |
| restriction | 7 |
| RESTRICTED | 6 |
| restrict | 3 |

### The capital letter carries meaning

Restricted Subsidiary 250 times, Restricted Payment 29: both defined terms. A stemmer would put them in one count of "restrict" with the ordinary word.

**The course keeps words whole. No stemming library is used.**

Source: Boyd Gaming Corporation, indenture for the 4.750% Senior Notes due 2031, Exhibit 4.1 to the 8-K filed 2021-06-08, accession 0001193125-21-185495.

# Excel side by side

### Excel side by side (tested in Excel)

| | |
|---|---|
| `=LEN(A2)` | 1,254, equal to len(text) |
| `=SEARCH("inflation",A4) and =FIND("inflation",A4)` | 195 and 336: Python + 1 |
| `=(LEN(A2)-LEN(SUBSTITUTE(LOWER(A2),"inflation","")))/LEN("inflation")` | 2; tokenize: 1 |
| `=TEXTBEFORE(TEXTAFTER(A3,"percentage point to "),"  percent")` | 3-3/4 to 4 |
| `=ROWS(TEXTSPLIT(A3,," "))` | 30; tokens 32 |
| `=REGEXEXTRACT(A3,"[0-9/-]+ to [0-9/-]+ percent")` | 3-3/4 to 4 percent |
| `=REGEXEXTRACT(A4,"[0-9/-]+ to [0-9/-]+ percent")` | #N/A on U+2011 |
| `=REGEXEXTRACT(SUBSTITUTE(A4,UNICHAR(8209),"-"),"[0-9/-]+ to [0-9/-]+ percent")` | 3-1/2 to 3-3/4 percent |
| `=WORDCOUNT(A2)` | #NAME? |

**No tokenizer in Excel. A cell keeps 32,767 characters: one paragraph per row (1,714 rows).**

# Save the day: start textkit

### 1. Init

The work folder, local settings, the guard

### 2. .gitignore

adds .env and cache/ from day 1

### 3. Commit

mlkit beside textkit and its tests

### 4. Status

the data copies stay untracked

```
 .gitignore      |   5 +
mlkit.py         | 568 ++++++
```

# The AI check

The docstring is the prompt. The code that comes back runs, and its number looks fine.

1

## Docstring

Units, conventions, and what the function must never do: what you asked for.

→

2

## Code back

Written for the course in the style of an AI answer. Read it.

→

3

## Test first

Assert what the docstring promises. Run it: it fails.

→

4

## Fix

Find the rule it broke, fix it, run the test again.

**Today: the splitter never normalized U+2011: "longer" 393 times, plus 7 odd one-word "longer-term".**

*Code that runs is not the same as code that is right.*

# What you can do now

- Treat a document as a string; a substring is not a word
- Make look-alike characters plain with normalize_text
- Pull the target range out of every statement, checked
- Find an indenture's sections and its defined terms
- Do the same string work in Excel, with the same traps

**Week 1, Day 2** Bag of words and TF-IDF